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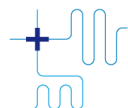
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ABSTRACT

For precise measurement of the moment of inertia (MOI) of asymmetric objects, using an air bearing torsional pendulum, the nonlinearity of the torsion bar and the damping effect caused by the interaction between air and sample need to be included in the analysis. The instantaneous undamped natural frequency and instantaneous damping coefficient of the motion are obtained from the Hilbert transform of the angular displacement signal and are used for data fitting to calculate the MOI. Measurements for algorithm verification were made using a special sample with large windvanes to increase the damping effect. The nonlinear analysis of the MOI experimental results indicated a relative error of less than 0.1% compared with 0.3% for classical linear analysis of the data.

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I. INTRODUCTION

Various mass properties of rigid bodies are commonly defined, such as the object's mass, location of the center of gravity, and moment of inertia (MOI).^{1–3} The MOI is a physical quantity that represents the inertia of an object in the process of rotation.^{4,5} It is often necessary to measure the MOI of an object in applications relating to aerospace, automobiles, machinery, instrumentation, and other industrial fields. For example, in the automotive field, the MOI is a key parameter in analyzing vehicle performance through dynamic modeling, potentially affecting the safety and stability of the vehicle.^{6,7} In the aerospace field, the MOI affects the running orbit, attitude, and flight stability of aircraft, providing an important basis for the design, analysis, and optimization of aircraft shape.^{8,9} Therefore, accurate measurements of the MOI have important practical significance.

Measuring the MOI requires a dynamic procedure, such as simple rotations around an axis fixed in space, complex motions, or

vibrations. Small rotational velocities, angles, or torques can be measured from the motion of the body and then used to derive the MOI using a suitable mathematical model.^{10,12} Using the equations of motion or their simplifications, these dynamic methods can be subdivided into three classes according to the introduced measurement constraints: methods with no limitation on rigid body motions, methods based on small angular motions, and methods based on very small angular motions or small rotations.^{13–17} Existing MOI measurement methods include the frequency response, compound pendulum, line pendulum, and torsion pendulum methods.¹¹ Tang developed an improved trifilar torsional pendulum for the experimental determination of the center of gravity and inertia tensor (the three MOIs and the three products of inertia) for an irregular-shaped body, reducing the relative error in the determination of the MOIs to less than 1%.¹⁸ Previati measured the MOI of rigid bodies by means of multi-filar pendulums and investigated the effect of deformations in the testing structure on the measurement of relatively large or heavy bodies both numerically and experimentally.¹⁹

Furthermore, Liu proposed a method for measuring the inertia properties of a rigid body based on a 3-URU parallel mechanism, which is a passive method relying on the gravitational field that does not require any excitation device.²⁰ Nripen established a torsion rod and flexure model for inertia measurements and proposed an MOI calculation method based on the frequency of free torsional oscillations.²¹ Based on the rigid-body complete motion equation, Brancati proposed a least-squares optimization procedure using experimental data acquired from static and dynamic tests.²² Hou presented an improved trifilar pendulum method for measuring the inertia parameters of odd-shaped bodies, in which the angle measurement process can be omitted.²³ Based on a dynamic balancing model, Bacaro proposed an MOI method for space payloads, allowing all ten inertia parameters to be measured from the torques and angular velocities generated by rigidly rotating the specimen three times.²⁴ Olmedo developed a set of MOI measuring devices based on a torsion string for a small off-the-shelf quadrotor unmanned aerial vehicle, the angular position of which can be obtained by a high-resolution quadrature optical encoder, and investigated the effects of viscous and Coulomb friction.²⁵ Farsoni presented a procedure for identifying the inertial parameters of the payload mounted on the end-effector of a robotic manipulator operating in a collaborative setup.²⁶

In the frequency response MOI measurement method, the measured object is hung from a flexible rubber rope while multiple parameters are calculated, resulting in limited bearing capacity and low measurement accuracy. For the complex pendulum method, the swing angle of the object affects the MOI measurement accuracy, and large objects create certain safety hazards. For the line pendulum method, the bearing capacity of the cycloid is limited, and the tension effect of the cycloid affects the MOI measurement accuracy.²⁷ The torsion pendulum method is widely used and has the advantages of high measurement accuracy, large bearing capacity, and simple operation. The MOI measuring principle of the torsion pendulum method is based on the functional relationship between the motion characteristics of the object placed on the torsion pendulum and the MOI. One of the basic components of a torsion pendulum is an air-floating turntable. Unlike a mechanical turntable, this is not influenced by dry friction. Its theoretical basis is the linear torsion pendulum motion, which allows the MOI to be calculated according to the period and amplitude ratio of the torsion pendulum.²⁸ However, as the sample dimensions increase, nonlinear factors have a greater influence on the MOI measurements, such as the nonlinear air damping.⁶ The motion characteristics of a nonlinear torsion pendulum are quite different from those of a linear torsion pendulum. For linear torsional pendulum motion systems, the amplitude ratio and period of torsional pendulum motion are fixed. However, for nonlinear torsional pendulum motion systems, the amplitude ratio decreases continuously under the effect of nonlinear damping. The period of the torsion pendulum changes with the amplitude due to the nonlinear restoring torques. These factors mean that existing linear methods cannot accurately describe the actual torsion pendulum motion.^{29–33} Meanwhile, the period solutions to the equation of motion for both duffing oscillators and compound pendula with quadratic damping are approximate, which would introduce the calculation error of the moment of inertia.³⁴

To enable high-precision MOI measurements in asymmetric objects, this paper describes a nonlinear measurement model based on an air bearing table. The influences of the restoring and damping torques on the motion of the torsion pendulum are analyzed, and the effectiveness of the measurement model and calculation method is verified experimentally.

II. MOI MEASUREMENT BASED ON A NONLINEAR TORSION PENDULUM

A. Traditional linear mathematical model of a torsion pendulum

Figure 1 shows the principle of the torsion pendulum method and the corresponding design structure of the measurement system. The system consists of an air floating turntable, an excitation device, a locking positioning device, a grating sensing system, a flexure hinge, and an air distribution system. During MOI measurements, the test object is fixed on the horizontal torsion table through a fixing mount. One end of the torsion bar is connected with the horizontal torsion table and the other is attached to the flexure hinge. After excitation, free torsional pendulum movement occurs under the action of the restoring torque of the torsion bar. The MOI J of the sample plus the air-floating turntable about the central vertical axis is a fixed value independent of time.

The excitation device, shown in Fig. 2(a), consists of a connecting rod, an adjusting screw, and a cylinder. The connecting rod is fixed on the mounting plate of the air-floating turntable. Extension of the cylinder provides an initial torsion angle for the torsion pendulum movement of the air-floating turntable through the connecting rod. Before conducting measurements, this initial angle can be changed by limiting the moving distance of the connecting rod with the adjusting screw. The adjusting screw needs to be fixed to prevent interference with the torsional pendulum movement during the measurements. The locking positioning device can be used to stop the torsion pendulum movement of the turntable and return it to its initial position through the expansion and contraction of the cylinder, as shown in Fig. 2(b).

The grating sensing system includes the scale grating, grating reading head, and digital display meter. The type of the grating ruler is RENISHAW RGS40-S, which has a grating spacing of $40\text{ }\mu\text{m}$. The principle of angular displacement measurement is shown in Fig. 3. Compared with laser reflection, this has the advantages of higher integration, smaller volume, and less interference from the external environment. The scale grating is fixed at the edge of the air-floating turntable, whose radius R is 325 mm , and the output of the grating is processed by a digital display meter to obtain the arc displacement l of the air-floating turntable. The corresponding angular displacement signal θ can then be obtained from the arc displacement l of the air floating turntable as

$$\theta = l/R. \quad (1)$$

As shown in Fig. 4, the traditional linear measurement model converts the digital angular signal θ into a series of square wave signals through the switch in the software. For a linear system, the torsion pendulum period T is equal to the average period of the square wave,

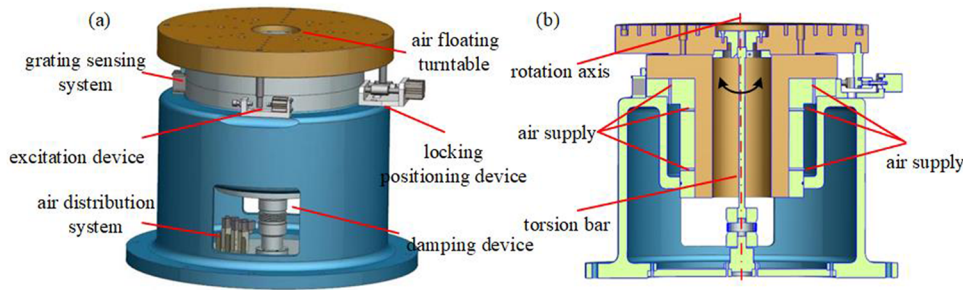


FIG. 1. (a) The structure of the measurement system and (b) the cross section showing the vertical torsion bar and the air floating turntable.

$$T = \frac{T_1 + T_2 + \dots + T_n}{n}. \quad (2)$$

$$J = \frac{K}{(2\pi)^2} T^2, \quad (3)$$

The MOI J of the sample plus the air-floating turntable about the central vertical axis can be calculated according to the torsional pendulum trajectory as^{28,35}

where K is the linear stiffness coefficient of the torsion bar, which is determined by the material properties and is assumed to be a linear time-invariant value, and T is the torsional pendulum period.

The traditional linear measurement model assumes that the torsion pendulum motion is constant, that is, its period and amplitude are basically unchanged. However, this is not always the case, especially for the measurement of large-scale samples. In addition, the traditional model only uses a few points in the angular displacement curve, which could make the measurement results vulnerable to noise interference.

B. Nonlinear mathematical model of a torsion pendulum

The nonlinear measurement model can be applied to the MOI measurement system (Fig. 1). As a single degree of freedom system, using the Lagrange equation, its kinematics can be expressed as

$$\frac{d}{dt} \left(\frac{\partial E}{\partial \dot{\theta}} \right) - \frac{\partial E}{\partial \theta} + \frac{\partial U}{\partial \theta} = Q, \quad (4)$$

where E and U represent the kinetic energy and potential energy of the system, respectively, θ is the torsion pendulum angle, and Q is a generalized torque, including damping and external excitation.

When the mass characteristics of the measured object are constant, the kinetic energy of the MOI system is as follows:

$$E = 0.5 \times J \times \dot{\theta}^2, \quad (5)$$

where J is the total MOI in the turntable coordinate system.

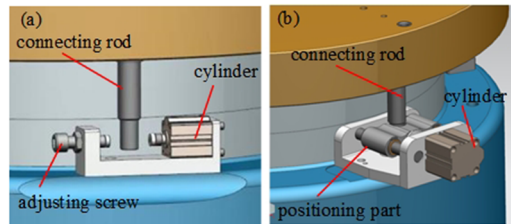


FIG. 2. Composition of the excitation device (a) and locking positioning device (b).

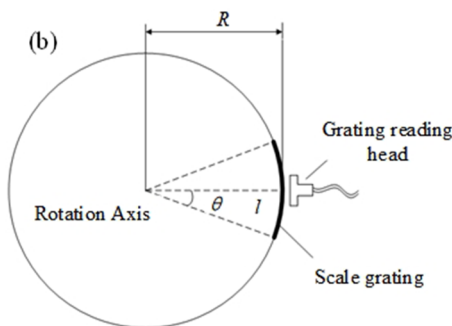
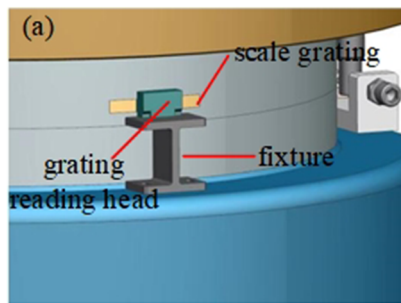


FIG. 3. Grating sensing system (a) and its principle of angle measurement (b).

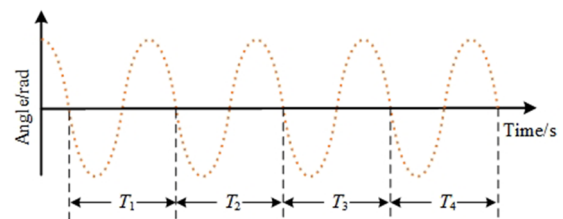


FIG. 4. Measurement principle of the torsional pendulum period for an undamped linear system.

Due to the influence of the elastic torsion bar on the air-floating turntable, the restoring torque M of the system is equivalent to a nonlinear torque, such as

$$M = K_1\theta + K_3\theta^3, \quad (6)$$

where K_1 is the linear stiffness coefficient and K_3 is the nonlinear stiffness coefficient of the torsion bar.

The system potential energy U can be expressed as

$$U = \int_0^\theta M d\theta = \frac{1}{2}K_1\theta^2 + \frac{1}{4}K_3\theta^4. \quad (7)$$

The air film is not influenced by dry friction, so the generalized moment Q of the system only needs to consider the nonlinear air damping,

$$Q = -C_1\dot{\theta} - C_2\dot{\theta}|\dot{\theta}|, \quad (8)$$

where C_1 and C_2 are the linear and quadratic damping torque coefficients, respectively.

By substituting Eqs. (5), (7), and (8) into Eq. (4), the MOI measurement model can be derived as

$$J\ddot{\theta} + C_1\dot{\theta} + C_2\dot{\theta}|\dot{\theta}| + K_1\theta + K_3\theta^3 = 0. \quad (9)$$

The normalized nonlinear MOI model is

$$\ddot{\theta} + c_1\dot{\theta} + c_2\dot{\theta}|\dot{\theta}| + k_1\theta + k_3\theta^3 = 0, \quad (10)$$

where $c_1 = C_1/J$, $c_2 = C_2/J$, $k_1 = K_1/J$, and $k_3 = K_3/J$.

The expression of the MOI model solution can be derived,^{36,37} but the unknown parameters usually require multiple iterations to obtain. Since the stiffness coefficient of the torsion bar is independent of the sample, only the normalization coefficients k_1 or k_3 needs to be determined using the angular displacement signal for MOI calculation.

C. Calculation process for a nonlinear torsion pendulum

For convenience of analysis, the torsional pendulum movement can be represented by the skeleton linear model,³⁸

$$\ddot{\theta} + 2h_0\dot{\theta} + \omega_0^2\theta = 0, \quad (11)$$

where h_0 is the instantaneous damping coefficient and ω_0 is the instantaneous undamped natural frequency, which can be derived as³⁹

$$\omega_0 = [\omega^2 - \ddot{A}/A + 2\dot{A}^2/A^2 + \dot{A}\dot{\omega}/A\omega]^{1/2}, \quad (12)$$

where A is the instantaneous amplitude and ω is the instantaneous angular frequency.

The instantaneous damping coefficient h_0 can be derived as

$$h_0 = -\dot{A}/A - 0.5\dot{\omega}/\omega. \quad (13)$$

These instantaneous parameters could be obtained by Hilbert transform. As a time-frequency analysis method, Hilbert transform

has the characteristics of self-adaptation, high resolution, and clear physical meaning, which is applied to parameter estimation for nonlinear systems, such as nonlinear vibration, multi degree of freedom motion, and other fields. Then, the instantaneous amplitude A can be written as³⁸

$$A = \sqrt{\theta^2 + \left| \int_{-\infty}^{+\infty} \frac{\theta(\tau)}{t - \tau} d\tau \right|^2}. \quad (14)$$

The instantaneous angular frequency is given by

$$\omega = d \left[\arctan \left(\int_{-\infty}^{+\infty} \frac{\theta(\tau)}{t - \tau} d\tau / \theta \right) \right] / dt. \quad (15)$$

According to Ref. 38, the normalized MOI measurement model can be expressed as

$$\ddot{\theta} + \sum_{j=0}^n c_j |\dot{\theta}|^j \cdot \text{sgn}(\dot{\theta}) + \sum_{i=1}^m k_i |\theta|^i \cdot \text{sgn}(\theta) = 0. \quad (16)$$

The instantaneous undamped natural frequency and the instantaneous damping coefficient can be derived as

$$\omega_0(A) = \left[\sum_{i=1}^m \frac{2}{\sqrt{\pi}} \frac{\Gamma(\frac{i}{2} + 1)}{\Gamma(\frac{i+1}{2} + 1)} k_i \cdot A^{i-1} \right]^{\frac{1}{2}}, \quad (17)$$

$$h_0(A\omega) = \sum_{j=0}^n \frac{1}{\sqrt{\pi}} \frac{\Gamma(\frac{j}{2} + 1)}{\Gamma(\frac{j+1}{2} + 1)} c_j \cdot (A\omega)^{j-1}, \quad (18)$$

where Γ is the gamma function. Several terms of interest are

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}, \Gamma(2) = 1, \Gamma(3) = 2, \Gamma\left(\frac{5}{2}\right) = \frac{3\sqrt{\pi}}{4}, \dots$$

By comparing Eqs. (10) and (16), the relationship between the instantaneous undamped natural frequency, the instantaneous damping coefficient, and the normalized measurement model can be derived as

$$\omega_0(A) \approx \left[k_1 + \frac{3}{2} k_3 A^2 \right]^{\frac{1}{2}}, \quad (19)$$

$$h_0(A\omega) \approx \left[\frac{c_1}{2} + \frac{4c_2}{3\pi} A\omega \right]^{\frac{1}{2}}. \quad (20)$$

The parameters in Eq. (19) can be calculated from the polynomial fitting of the instantaneous undamped natural frequency. The relationship between the nonlinear MOI and the instantaneous undamped natural frequency is

$$J = K_1/k_1. \quad (21)$$

In summary, the proposed MOI calculation process is as follows: First, to remove the interference elimination, the angular displacement signal of the measured object is decomposed and filtered using EMD. Second, the instantaneous amplitude and instantaneous angular frequency can be constructed from EMD results by Hilbert transform. Then, the above information can be used to calculate

the instantaneous undamped natural frequency and instantaneous damping coefficient according to Eqs. (12) and (13). Therefore, using the least square curve fitting method, the parameter values of the MOI measurement model are estimated according to Eq. (19). Finally, the MOI of the measured object can be obtained according to the stiffness coefficient of the torsion bar.

III. SIMULATION AND ANALYSIS OF MOI ALGORITHM

For the MOI calculation method based on an air bearing platform, the results are related to factors such as the air damping and nonlinear restoring torque. The purpose of this section is to use the established mathematical model to analyze the influence of these two parameters on the torsional pendulum motion, as this will provide theoretical guidance for analyzing the relationship between the moment of inertia and the motion characteristics of the nonlinear torsion pendulum.

The parameter values of the simulation signals under nonlinear conditions are listed in Table I. The initial value of the angular displacement is 0.06 rad, the initial angular velocity is zero, and the sampling frequency is 2000 Hz.

A. Influence of nonlinear air damping on torsional pendulum motion

For an object with a constant MOI, the amplitude of the torsional pendulum is mainly related to the air damping torque coefficients C_1 and C_2 . To analyze their influence on the motion of the torsion pendulum, the amplitude ratio of the angular displacement signal, representing the amplitude ratio of adjacent periods, is compared under the condition of linear or nonlinear air damping, which reflects the attenuation degree of the signal amplitude.

In the undamped state, $C_1 = C_2 = 0$, the amplitude of the angular displacement signal is not attenuated. When $C_1 \neq 0$, $C_2 = 0$, and $K_3 = 0$, the MOI model is in the condition of linear air damping for which the amplitude ratio is constant and the amplitude decays exponentially. Under the condition of nonlinear air damping ($C_1 \neq 0$, $C_2 \neq 0$), the amplitude ratio decreases nonlinearly, as shown in Fig. 5. Therefore, when there is nonlinear air damping in the measurement system, the amplitude ratio of the angular signal will decrease continuously.

B. Calculated influence of nonlinear restoring torque on the torsion pendulum motion

For an object with a constant MOI, the period of motion for the torsion pendulum is related to the stiffness coefficients, i.e., K_1 and K_3 . To analyze the influence of the nonlinear restoring torque on

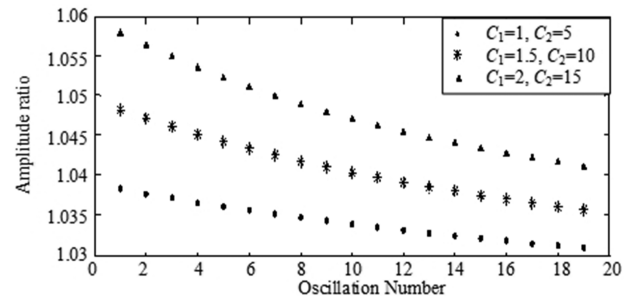


FIG. 5. Amplitude ratios of simulation signals for nonlinear damping.

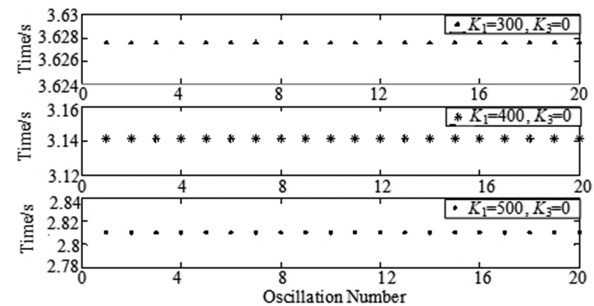


FIG. 6. Periods of simulation signals for linear restoring force.

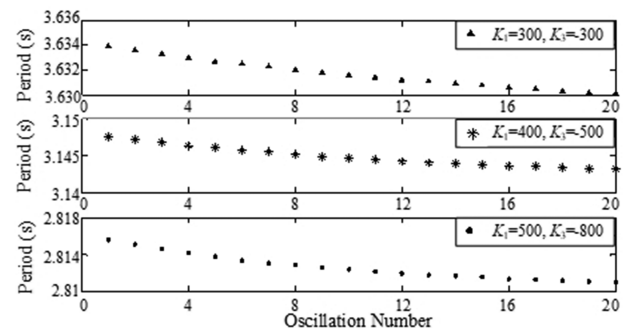


FIG. 7. Periods of simulation signals for nonlinear restoring force.

the motion of the torsion pendulum, the period is compared under linear and nonlinear restoring torques.

Under a linear restoring torque ($K_1 \neq 0$, $K_3 = 0$), the real-time period of the simulation signal is shown in Fig. 6. It can be seen that

TABLE I. Parameter values of MOI simulation signals.

No.	J (kg m ²)	C_1 (kg m ² s ⁻¹)	C_2 (kg m ² rad ⁻¹)	K_1 (N m rad ⁻¹)	K_3 (N m rad ⁻³)
1	100	1	5	300	-300
2	100	1.5	10	400	-500
3	100	2	15	500	-800

with only a linear restoring torque, the time interval of each cycle of the angular displacement signal is constant. The values of C_1 and C_2 are shown in Table I.

Under the condition of a nonlinear restoring force ($K_1 \neq 0$, $K_3 < 0$), the period of the simulation signal is shown in Fig. 7. The time interval of each cycle of the angular displacement simulation signal is continuously decreasing in the circumstances. This indicates that the nonlinear restoring torque of the measurement system changes the signal period of angular displacement.

IV. RESULTS OF CALIBRATION TESTS

Experiments were conducted in which a loading platform, including a strip stage and eight chassis, was fixed to the air-floating turntable. The positional relationship between the chassis and the coordinate system was known. The accuracy of the model under different conditions was verified by placing the samples on the different chassis (see Fig. 8).

A. Calibration of stiffness coefficient of torsion bar

The torsion bar (made of 60Si2Mn) has an effective length and radius of 505 and 10 mm, respectively, and a shear elastic modulus of $7.9231 \times 10^{10} \text{ N m}^{-2}$. According to Eq. (21), the torsion bar coefficient should be calibrated before the MOI measurements. Therefore, two cylindrical calibration masses, A and B, with known MOIs were introduced. The specific process was performed as follows.

First, the angular displacement signal of the torsional pendulum motion was measured under the no-load condition. Using the proposed MOI calculation process in Sec. II, the corresponding normalized torsion bar stiffness coefficients k_{11} and k_{13} were obtained according to Eqs. (19) and (20). Assuming that J_0 is the measured MOI in the idle condition, the torsion bar coefficients can be expressed as

$$k_{11} = K_1/J_0, \quad (22)$$

$$k_{13} = K_3/J_0, \quad (23)$$



FIG. 8. Calibration of torsion bar coefficient for the MOI system.

where K_1 and K_3 are the normalized torsion bar stiffness coefficients under the no-load condition, respectively.

Under the no-load condition, the normalized model parameter values are presented in Table II.

Second, cylindrical calibration masses A and B were fixed on the chassis of the strip stage, as shown in Fig. 8. Their MOIs J_A and J_B about their respective centerlines were calculated according to their overall dimensions. The MOI J_{Al} and J_{Bl} about the rotating shaft of the air floating turntable is

$$\begin{aligned} J_{Al} &= J_A + m_A l_A^2 = 28.65 \text{ kg m}^2, \\ J_{Bl} &= J_B + m_B l_B^2 = 28.65 \text{ kg m}^2, \end{aligned} \quad (24)$$

where m_A and m_B are the masses of A and B and l_A and l_B are the distances between the corresponding sample and the rotation center.

Similarly, the torsion bar coefficients under loading were calculated from the angular displacement signal as

$$k_{12} = K_1/(J_0 + J_{Al} + J_{Bl}), \quad (25)$$

$$k_{32} = K_3/(J_0 + J_{Al} + J_{Bl}). \quad (26)$$

Under the loading cylindrical calibration mass condition, the normalized measurement model parameter values are presented in Table III.

According to Eqs. (24)–(28), the torsion bar stiffness coefficients K_1 and K_3 and the MOI under the no-load condition J_0 can be derived as

$$K_1 = k_{11} \cdot k_{12} \cdot (J_{Al} + J_{Bl}) / (k_{11} - k_{12}), \quad (27)$$

$$K_3 = k_{31} \cdot k_{32} \cdot (J_{Al} + J_{Bl}) / (k_{11} - k_{12}), \quad (28)$$

$$J_0 = k_{12} \cdot (J_{Al} + J_{Bl}) / (k_{11} - k_{12}). \quad (29)$$

Five experiments were carried out using this procedure, and the average value was taken as the final calibration result. The calibrated values of the torsion bar stiffness coefficients K_1 and K_3 are 426.35 and $-511.62 \text{ N m rad}^{-3}$, respectively, and the calibrated MOI under the no-load condition is 94.50 kg m^2 (Table IV).

TABLE II. Model parameters with normalized MOI under the no-load condition.

Model parameter	c_{11} (s^{-1})	c_{21} (rad^{-1})	k_{11} ($\text{rad}^{-1} \text{ s}^{-2}$)	k_{31} ($\text{rad}^{-3} \text{ s}^{-2}$)
Measured value	0.0017	0.0150	4.5119	-5.4140

TABLE III. Model parameters with normalized MOI under the loading cylindrical calibration mass condition.

Model parameter	c_{12} (s^{-1})	c_{22} (rad^{-1})	k_{12} ($\text{rad}^{-1} \text{ s}^{-2}$)	k_{32} ($\text{rad}^{-3} \text{ s}^{-2}$)
Measured value	0.0012	0.0112	2.8088	-3.3721

TABLE IV. Calibration results of stiffness and J_0 for tare measurement.

Number	K_1 (N m rad ⁻¹)	K_3 (N m rad ⁻³)	J_0 (kg m ²)
1	426.37	-511.62	94.50
2	426.43	-511.75	94.52
3	426.30	-511.60	94.49
4	426.36	-511.57	94.49
5	426.29	-511.56	94.48
Average value	426.35	-511.62	94.50
Standard deviations	0.06	0.08	0.02

B. Test samples for the MOI measurement system

As shown in Fig. 9, two experimental samples, C and D, were installed on the MOI measurement system to validate the nonlinear model. The theoretical MOIs of the test samples about their own centerlines were accurately calculated with the advanced quality attribute analysis function of the computer-aided design software Unigraphics NX (UG).

The test samples consisted of a damping plate, support, and standard base. The damping plate was a rectangular rigid plastic plate of uniform density, which prevented the damping plate from shaking during the torsion pendulum process. A slot is cut into the support, and the plastic plate is fixed to it by screws. The base was made of iron cylinders of uniform density, and its lower part had a cylindrical groove that matched the chassis of the strip-shaped stage, allowing the test samples to be fixed in place. α is the angle between the damping plate and the perpendicular to the strip stage. When the damping plate is parallel to the strip stage, the angle α is 90°.

A coordinate measuring machine with accuracy $\pm 8 \mu\text{m}$ was used to measure each part of the samples, and a precision electronic balance with accuracy $\pm 1 \text{ g}$ was used to measure the mass of each part (see Table V). It should be noted that there are some hollow parts in the sample and the size of the fixing hole is not given.

Based on the geometric dimensions obtained from these measurements, three-dimensional models of the test samples were

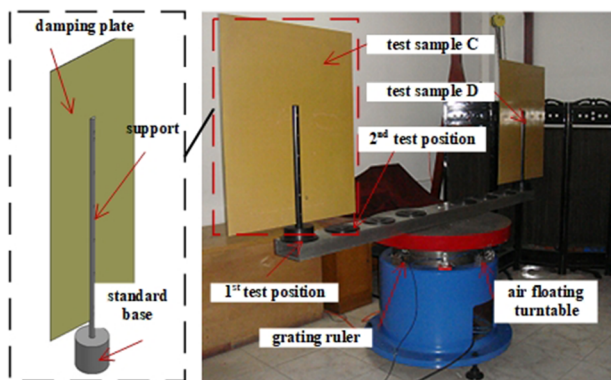
TABLE V. Overall dimensions and mass of the experimental samples.

Sample parts	Parameter	Test sample C	Test sample D
Damping plate	Length (mm)	800.035	800.032
	Height (mm)	1000.046	1000.041
	Width (mm)	2.503	2.506
	Mass (kg)	5.385	5.393
Support	Diameter (mm)	39.983	39.988
	Length (mm)	610.051	610.065
	Mass (kg)	5.408	5.410
Base	Diameter (mm)	155.053	155.062
	Length (mm)	58.033	58.037
	Mass (kg)	8.071	8.072
Distance from central axis to position 1 (mm)		1199.742	1199.364
Distance from central axis to position 2 (mm)		899.460	900.064

established using the UG software. The theoretical MOIs of the experimental samples about their own centerlines were calculated through the advanced quality attribute analysis function of UG. The theoretical MOIs of the experimental samples about the air-floating rotating shaft were then calculated using the parallel axis theorem. The results are presented in Table VI. The main sources of error include the mass, overall dimensions, and installation position. The standard uncertainties in the calculated MOIs of samples C and D are approximately equal to 0.013%. Meanwhile, due to the slow linear speed of the sample, the added mass effects of the air surrounding the sample have not been taken into account.

C. Angular displacement measurement results

To ensure that the measurement status was consistent with that of the calibration experiment, the MOI under the no-load condition J_0 was measured before the product measurements were conducted. This value is the total MOI of the air-floating turntable and strip

**FIG. 9.** Moment of inertia measurement experiment using samples C and D with $\alpha = 90^\circ$, which enhances the damping.**TABLE VI.** MOI theoretical values of experimental samples.

Parameter	Test sample C	Test sample D
Total mass (kg)	18.864	18.875
MOI theoretical value about centerline (kg m ²)	0.313	0.314
MOI theoretical value about rotation axis in position 1 (kg m ²)	27.47	27.49
MOI theoretical value about rotation axis in position 2 (kg m ²)	15.59	15.61

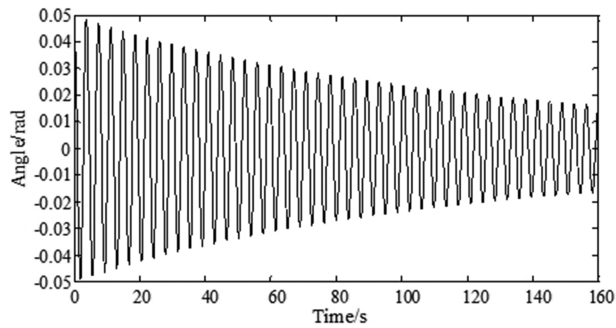


FIG. 10. Angle displacement signal of the first test position($\alpha = 90^\circ$).

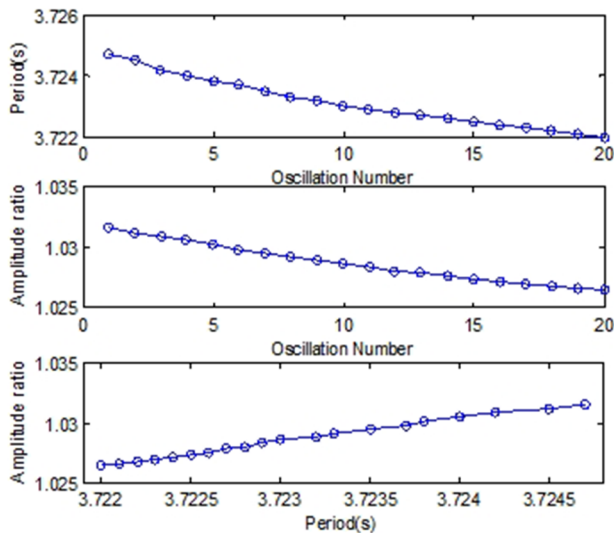


FIG. 11. Periods (a) and amplitude ratio (b) at the first test position($\alpha = 90^\circ$).

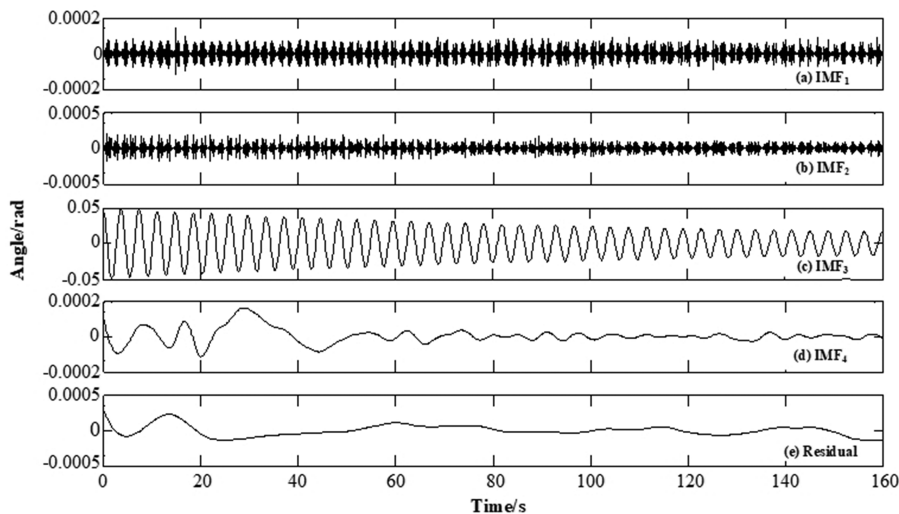


FIG. 12. EMD processing result of the experimental signal at the first test position ($\alpha = 90^\circ$), including IMF₁ (a), IMF₂ (b), IMF₃ (c), IMF₄ (d), and residuals (e).

stage. This step can be omitted on the premise that the equipment is calibrated regularly.

The second step involved fixing test samples C and D onto the chassis of the strip stage and measuring the corresponding angular displacement signal, as shown in Fig. 10. The measured MOI value J in the first test position with $\alpha = 90^\circ$ is the sum of the MOI of the air-floating turntable, the strip stage, and the experimental sample. The amplitude of the angular displacement signal gradually decreases, which indicates that there would be obvious errors when using the traditional periodic method to process this angle data.

Figure 11 shows the amplitude ratio and periods of the corresponding angular displacement signal. The gradual decrease of the period verifies the existence of a nonlinear restoring torque, while the amplitude ratio of the angular displacement signal verifies the existence of nonlinear air damping.

D. Empirical mode decomposition analysis

In the nonlinear MOI model, a very crucial step is to obtain its corresponding instantaneous angular frequency from the angular displacement signal. Empirical mode decomposition (EMD) is a time-frequency signal processing method in which signals are decomposed according to the timescale characteristics of the data without setting any basis functions in advance, namely, intrinsic mode functions (IMFs). IMFs can be considered as single-component signals, and their instantaneous angular frequency has a clear physical meaning. Therefore, before using the Hilbert transform to calculate the MOI, it is necessary to use EMD to decompose the angular displacement signal to adaptively remove the noise contained therein and facilitate the subsequent acquisition of smooth time-frequency signals.^{40–43}

Using the EMD algorithm, the angular displacement experimental signal was decomposed into several single-frequency sub-signals, as shown in Fig. 12. The processing results indicate that components IMF₁ and IMF₂ are high-frequency signals with small amplitudes that are generated by the vibration noise in the measure-

ment process. Therefore, these interference components need to be removed in the process of data calculation. IMF₃ is basically the same as the original signal. In addition, IMF₄ and the residuals contain low-frequency noise and nonlinear components. As their amplitude is very small and to obtain a smooth instantaneous frequency, these factors were ignored in the subsequent calculation process.

The filtered experimental signal IMF₃ was used to calculate the instantaneous undamped natural frequency and instantaneous damping coefficient using Eqs. (12) and (13). The results are shown in Fig. 13. To facilitate the subsequent calculation of the normalized measurement model parameters, their horizontal scales are different.

Using Eqs. (19) and (20), the two curves in Fig. 13 were fitted to give the least-squares curve, allowing the normalized measurement model parameter values of the first test position to be calculated. The parameter values are presented in Table VII.

Using the stiffness coefficient K_1 of the torsion bar and the MOI under the no-load condition J_0 , the measured values of the MOI for the two experimental samples about the rotating shaft were calculated according to

$$J = K_1/k_1 - J_0 = 55.00 \text{ kg m}^2. \quad (30)$$

From the theoretical values for the experimental samples, the relative error of the measured MOIs was calculated as

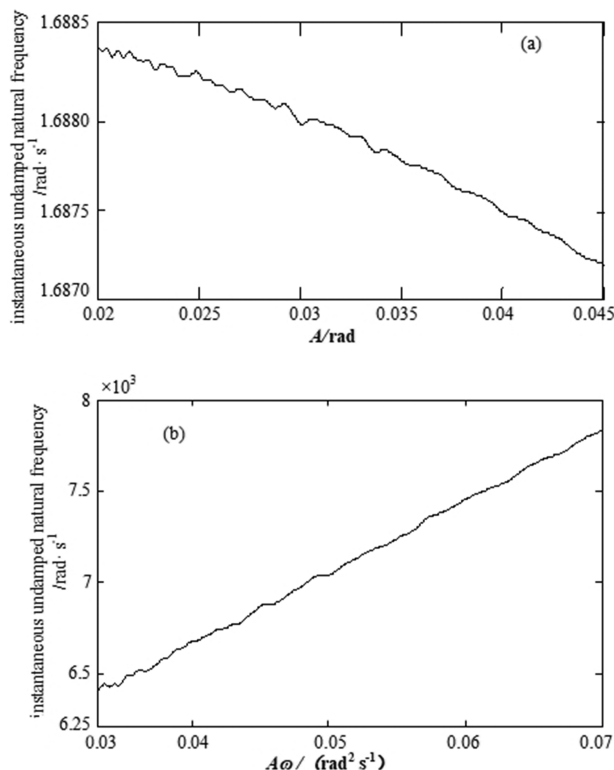


FIG. 13. Instantaneous undamped natural frequency (a) and instantaneous decay coefficient (b) at the first test position.

TABLE VII. Model parameters with normalized MOI for the experimental samples at the first test position ($\alpha = 90^\circ$).

Model parameter	$c_1 \text{ (s}^{-1}\text{)}$	$c_2 \text{ (rad}^{-1}\text{)}$	$k_1 \text{ (rad}^{-1} \text{ s}^{-2}\text{)}$	$k_3 \text{ (rad}^{-3} \text{ s}^{-2}\text{)}$
Measured value	0.0102	0.0921	2.8518	-3.4218

$$\delta = \frac{J - J_{Cl} - J_{Dl}}{J_{Cl} + J_{Dl}} \times 100\% = 0.07\%, \quad (31)$$

where J_{Cl} and J_{Dl} are the theoretical MOI values of samples C and D about the rotating shaft, respectively. Its standard experimental uncertainty is 0.1%.

E. MOI measurement results and comparison

The MOI experiments were conducted five times at the first test position ($\alpha = 90^\circ$) using the linear measurement model of Eq. (2) and the nonlinear measurement model of Eq. (19) using the MOI calculation method based on the Hilbert transform. The relative errors of the MOI measurements using these different measurement models are presented in Table VIII. The relative error of the MOI measurement based on the linear measurement model is large because the linear measurement model ignores the influence of nonlinear factors. The MOI measurement results based on the nonlinear measurement model are obviously better than those based on the linear measurement model, which verifies the effectiveness of the MOI measurement model based on the nonlinear dynamics.

A second set of measurements was made with the samples in position 2 ($\alpha = 90^\circ$), which is closer to the rotation center of the air flotation table (see Fig. 10). Thus, the damping torque is reduced. The results are given in Table IX. There is no obvious change in the measurement error of the proposed nonlinear analysis method, while the measurement accuracy of the linear model remains significantly affected by the air damping torque.

A third group of measurements was made with a different value of α at the first test position. The corresponding normalized measurement model parameter values are presented in Table X. As α increases, the air damping first increases and then decreases. The windward area of the damping plate is the largest when the damping plate is parallel to the strip stage ($\alpha = 90^\circ$), and so the effect of air damping is the most obvious.

TABLE VIII. Relative error of MOI for different measurement models at the first test position ($\alpha = 90^\circ$).

Number	Linear measurement model (%)	Nonlinear measurement model (%)
1	0.23	-0.06
2	0.32	0.04
3	0.35	0.08
4	0.25	-0.05
5	0.31	0.03
Average value	0.29 ± 0.05	0.01 ± 0.05

TABLE IX. Relative error of MOI for different measurement models at the second test position ($\alpha = 90^\circ$).

Number	Linear measurement model (%)	Nonlinear measurement model (%)
1	0.14	-0.07
2	0.23	0.05
3	0.18	-0.04
4	0.19	-0.03
5	0.25	0.06
Average value	0.20 ± 0.04	-0.006 ± 0.05

TABLE X. Model parameters with the normalized MOI for the experimental samples at the first test position with different angles α .

Angle α	c_1 (s^{-1})	c_2 (rad^{-1})	k_1 ($rad^{-1} s^{-2}$)	k_3 ($rad^{-3} s^{-2}$)
0°	0.0126	0.0077	2.8516	-3.4215
30°	0.0450	0.0422	2.8514	-3.4213
60°	0.0782	0.0731	2.8523	-3.4224
90°	0.0102	0.0921	2.8518	-3.4218
120°	0.0766	0.0734	2.8523	-3.4224
150°	0.0443	0.0423	2.8522	-3.4223
180°	0.0128	0.0074	2.8513	-3.4212

TABLE XI. Relative error of MOI for different measurement models at the first test position with different angles α .

Angle α	Linear measurement model (%)	Nonlinear measurement model (%)
0°	0.08	0.03
30°	0.14	0.05
60°	0.22	-0.04
90°	0.29	0.01
120°	0.25	-0.04
150°	0.16	-0.03
180°	0.11	0.06
Average value	0.18 ± 0.08	0.006 ± 0.05

Table XI indicates that the measurement error of the linear model is closely related to the change in damping. With a larger air damping torque, the measurement error of the linear method increases. However, the relative error of proposed method remains less than 0.1%, which is better than the linear measurement model. Moreover, the air damping moment does not affect the accuracy of the proposed method because the exact functional relationship between the MOI and the amplitude attenuation of the measured object is established through the instantaneous undamped natural frequency.

V. CONCLUSIONS

To solve the problem of obtaining high-precision MOI measurements for large-scale samples, this paper has proposed a new

nonlinear measurement model based on an air bearing torsion turntable. The proposed method compensates for the influence of air damping and the nonlinear restoring torque on the measurement results of the torsion pendulum method. The MOI can be calculated from the instantaneous undamped natural frequency of the torsional pendulum motion as derived from the angular displacement signal using a Hilbert transform. Experiments show that the MOI error of the proposed method is less than 0.1%, and the measurement accuracy and repeatability are better than those computed with the existing linear methods of analysis.

The air bearing turntable-based MOI measurement model described in this paper is different from the ball bearing turntable method, which must also consider dry friction. However, the content of this paper provides the framework for establishing a nonlinear mathematical model for the traditional MOI measurement methods based on ball bearing turntables.

In addition, existing measurement technology is mainly intended for samples with time-invariant MOIs, whereas MOIs of many measurement objects change with time. Thus, it is necessary to study MOI measurements of time varying objects, as these will be useful for unmanned aerial vehicles, aircraft, and underwater detectors in which the MOI varies with time due to structural transformations and fuel consumption.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jun Wang: Conceptualization (equal). **Liu Zheng:** Formal analysis (equal). **Yixu Wang:** Conceptualization (equal); Methodology (equal). **Na Miao:** Investigation (equal). **Xiaolin Zhang:** Methodology (equal); Validation (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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